Named Entity Recognition (NER) and Relation Extraction (RE) on ADKG and MDKG

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Abstract

Extraction from biomedical text remains challenging due to complex entity structures and long-range dependencies. In this work, we evaluate a span-based joint entity and relation extraction model (SpERT) on two PubMed-derived datasets, ADKG and MDKG. We first analyze dataset characteristics, noting class imbalance, and then assess model performance across entity and relation types. We further investigate cross-domain transfer learning by incorporating MDKG as auxiliary training data for ADKG. Results show that performance is dictated by high-frequency patterns but degrades on overlapping entities and long-distance relations. In addition, simple dataset merging does not improve the model in a cross-domain transfer learning setting.

1 Introduction

Named Entity Recognition (NER) and Relation Extraction (RE) can be used to identify structures in biomedical text. In this work, we apply a span-based model SpERT to two PubMed-derived datasets: ADKG, focused on Alzheimer's disease, and MDKG, which covers a broader range of mental disorders. We first perform exploratory data analysis on the entity and relation distributions before applying the SpERT model. We then investigate cross-domain transfer learning by incorporating MDKG as auxiliary training data for ADKG. Finally, we examine model performance through error analysis, focusing on overlapping entities, long-distance dependencies, and annotation inconsistencies.

This paper is organized as follows. Section 2 introduces the dataset. Section 3 describes the data, SpERT architecture, and cross-domain setup. Section 4 presents experimental results, including exploratory data analysis, performance comparisons, and error analysis. Finally, we discuss in Section 5 key findings and conclude the paper.

2 Data

We use two annotated datasets derived from PubMed abstracts: Alzheimer's Disease Knowledge Grap (ADKG) and Mental Disorder Knowledge Graph (MDKG). ADKG contains 8,031 sentences associated with Alzheimer's disease and related neurodegenerative conditions, and MDKG includes 6,678 sentences covering mental disorders such as schizophrenia, depression, bipolar disorder, PTSD, and OCD. Both datasets are annotated with biomedical entities and relations, with some overlap. The exact entities and relation can be found below.

https://www.notion.so/Datasets-Description-33b3d297a55f80879988c3ffbe3b03ca

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3 Methods

3.1 Data

The training, development, and test splits can be found above. The datasets were provided in JSON format with training, development, and test splits predefined. JSON Entity and relation type dictionaries were provided as well. For the cross-domain improvement, we concatenated the ADKG and MDKG train and dev splits, and merged their types. We performed preprocessing to ensure compatibility with the SpERT input format.

3.2 Span-based Entity and Relation Transformer (SpERT)

We adopt the Span-based Entity and Relation Transformer (SpERT) for NER and RE on ADKG and MDKG. SpERT uses a lightweight reasoning on BERT embeddings for entity recognition and a localized, marker-free context representation for relation classification.

We utilize the official implementation of SpERT [Eberts and Ulges, 2019]. The repository is provided below. We implement SpERT exactly as given in the repository. No changes were made.

`https://github.com/lavis-nlp/spert`

4 Experimental Results

4.1 Exploratory Data Analysis

We analyze and compare the entity type distributions in ADKG and MDKG in Figure 1. We also analyze the relation type distirbutions of ADKG and MDKG in Figure 2.

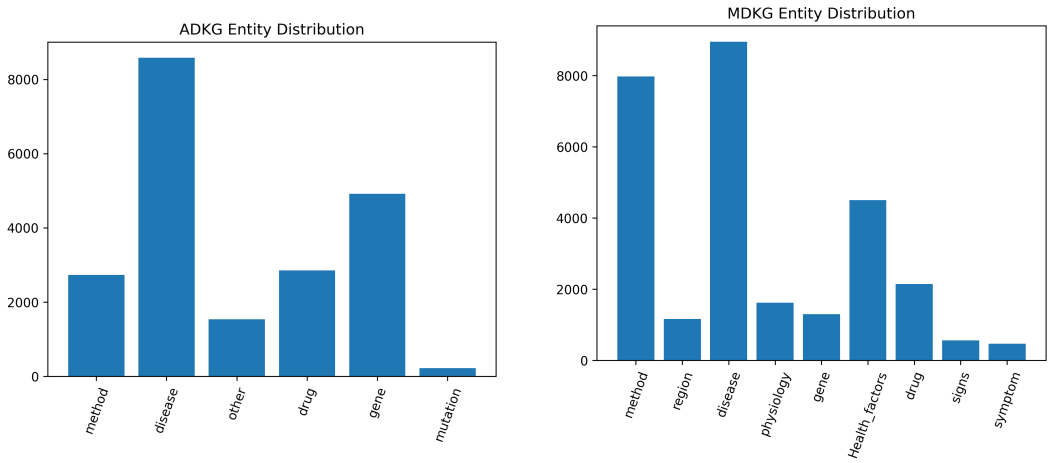

Figure 1: Entity type distributions in ADKG (left) and MDKG (right).

The distributions reveal many differences between the two datasets. First, MDKG has a more diverse pool of entities and relations, as it covers a broader coverage of mental health conditions. Second, in both ADKG and MDKG, a small number of entities and relations dominate, indicating class imbalance. Specifically, in ADKG entities, disease dominates, but mutation and other do not. In MDKG entities, method and disease, but signs and sympton do not. In ADKG relations, abbreviation_for dominates, but treatment_target_for does not. In MDKG relations, associated_with domminates, but located_in does not.

4.2 Model Results

We use the parameters given in the repository tutorial.

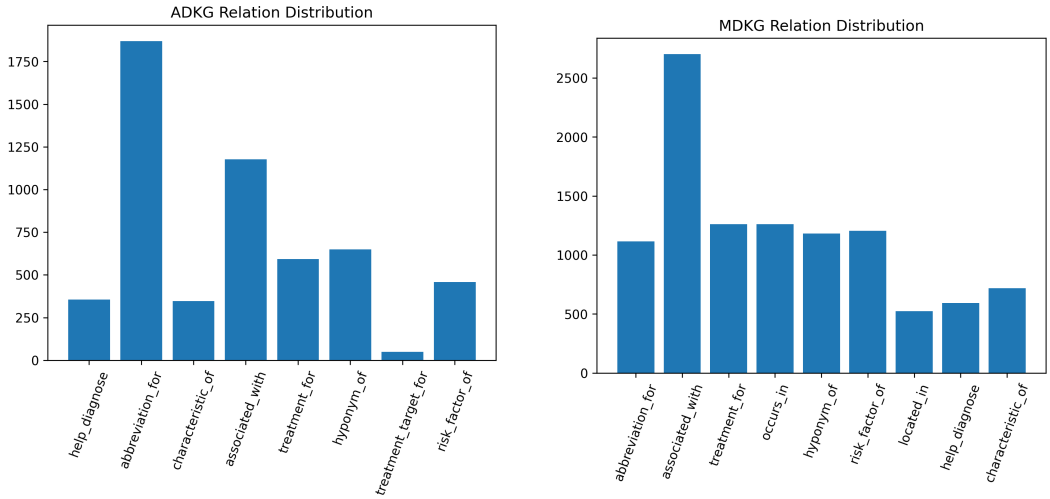

Figure 2: Relation type distributions in ADKG (left) and MDKG (right).

An entity is considered correct if the entity type and span is predicted correctly. A relation is considered correct if the relation type and the two related entities are predicted correctly (in span and entity type). Overall results are shown in Table 1. MDKG consistently achives higher F1 scores than ADKG, suggesting better model performance. Across both ADKG and MDKG, the micro F1 is higher than the macro F1.

Table 1: Micro and Macro performance across ADKG and MDKG for NER and RE.

Dataset	Avg	NER			RE		
		Precision	Recall	F1	Precision	Recall	F1
ADKG	Micro	71.64	64.61	67.95	45.27	37.97	41.30
	Macro	65.29	58.32	61.26	31.88	25.76	27.83
MDKG	Micro	80.96	79.80	80.38	53.77	49.86	51.74
	Macro	71.49	71.18	71.26	55.04	51.71	53.03

Table 2 reports the NER performance by entity on ADKG and Table 3 reports RE by relation on ADKG. For ADKG, NER had high F1 scores > 70 for disease (81.27) and mutation (71.58) and low scores < 50 for other (36.15) and method (49.47). RE had a high F1 score > 70 for abbreviation_for but low scores < 50 for all other relations.

Table 4 reports the NER performance by entity on MDKG and Table 5 reports RE by relation on MDKG. For MDKG, NER had high F1 scores > 70 for all entities except physiology, signs, and symptons and low scores < 50 for sympton (36.36). RE had a high F1 score > 70 for abbreviation_for but low scores associated_with (38.84) and characteristic_of (32.37).

Table 2: NER on ADKG by Entity.

Entity Type	Precision	Recall	F1
disease	81.48	81.07	81.27
gene	68.26	63.38	65.73
drug	75.86	54.43	63.39
method	53.11	46.31	49.47
mutation	69.39	73.91	71.58
other	43.66	30.85	36.15

Table 3: RE on ADKG by Relation.

Relation Type	Precision	Recall	F1
abbreviation_for	69.60	70.90	70.24
hyponym_of	40.82	43.01	41.88
characteristic_of	36.84	14.58	20.90
treatment_for	31.15	24.05	27.14
risk_factor_of	22.64	20.34	21.43
associated_with	15.89	10.56	12.69
help_diagnose	21.43	11.54	15.00
treatment_target_for	16.67	11.11	13.33

Table 4: NER on MDKG by Entity.

Entity Type	Precision	Recall	F1
disease	89.94	90.96	90.45
gene	73.56	71.50	72.51
drug	75.76	80.65	78.12
method	81.84	77.99	79.87
physiology	61.20	62.96	62.07
health_factors	80.63	75.26	77.85
region	72.08	81.02	76.29
signs	70.91	65.00	67.83
symptom	37.50	35.29	36.36

Table 5: RE on MDKG by Relation.

Relation Type	Precision	Recall	F1
abbreviation_for	74.32	73.83	74.07
associated_with	41.85	36.24	38.84
characteristic_of	41.79	26.42	32.37
occurs_in	50.60	59.86	54.84
help_diagnose	55.00	46.48	50.38
hyponym_of	66.24	66.67	66.45
risk_factor_of	58.21	49.06	53.24
treatment_for	54.61	56.85	55.70
located_in	52.78	50.00	51.35

4.3 Error Analysis

Overlapping Entities Due the variability in F1 scores, we investigate whether performance gaps arise from overlapping entities. Table 6 shows the error rates for overlapping and non-overlapping entities. The error rate is extremely high when there is overlap in entities, 0.94 and 1.0 for ADKG and MDKG respectively. The error rate is substantially lower when there is no overlap, 0.69 and 0.56 respectively.

Table 6: Error Rates for Overlapping vs Non-overlapping Entities.

Dataset	Overlap Error	Non-Overlap Error
ADKG	0.94	0.69
MDKG	1.00	0.56

Table 7: Average Span Distance for Correct vs Incorrect Relations.

Dataset	Distance (True)	Distance (False)
ADKG	0.65	7.10
MDKG	1.46	8.53

Long-Range Dependencies We also investigate if performance gaps arise from long-range dependencies. Table 7 shows the average span distance for correct and incorrect relations. The average distance for a correct relation is 0.65 tokens and 1.46 tokens for ADKG and MDKG respectively. The average distance for an incorrect relation is 7.10 tokens and 8.53 tokens for ADKG and MDKG respectively. This is supported by Figure 3, which shows that as relation distance increases, error rate also increases.

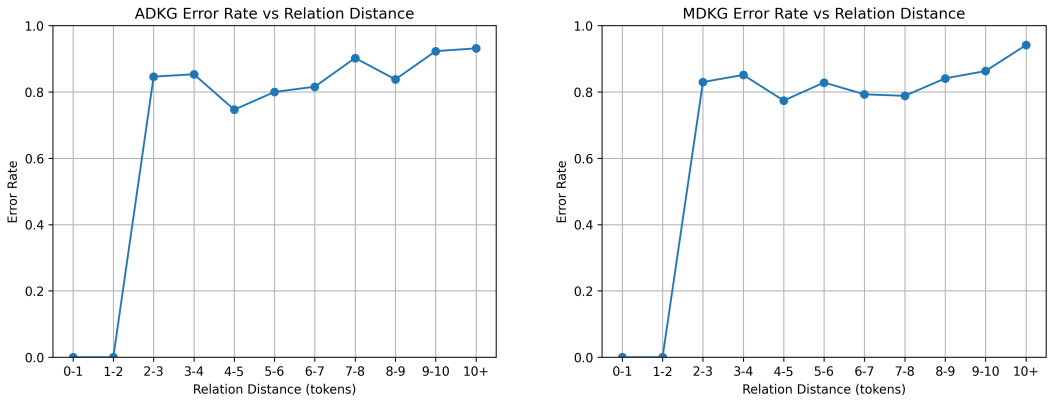

Figure 3: Error Rate vs Relation Distance for ADKG (left) and MDKG (right).

Edge Cases There were also several edge cases, where the relation appears valid but was not in the original annotations.

We examine Sample A from ADKG, where the original text is:

"Alzheimer 's disease (AD) is considered to be the most common cause of dementia and is an incurable , progressive neurodegenerative disorder ."

From 8, we see that the model predicts an additional relation that appears valid given the sentence context since AD is an abbreviatin for Alzeheimer's Disease. However, this relation is not present in the ground-truth annotations, likely due to inconsistencies in tokenization and uneven spacing during dataset construction, which may have led to it being overlooked during annotation.

True Entities
{type: disease, start: 0, end: 3} {type: disease, start: 4, end: 5} {type: disease, start: 15, end: 16} {type: disease, start: 22, end: 24}
True Relations
{type: risk_factor_of, head: 0, tail: 2} {type: hyponym_of, head: 0, tail: 3}

Predicted Entities
{type: disease, start: 0, end: 3} {type: disease, start: 4, end: 5} {type: disease, start: 15, end: 16} {type: disease, start: 22, end: 24}
Predicted Relations
{type: risk_factor_of, head: 0, tail: 2} {type: hyponym_of, head: 0, tail: 3} {type: abbreviation_for, head: 1, tail: 0}

Table 8: True vs predicted entities and relations for Sample A.

We examine another sample Sample B from MDKG.

"Musculoskeletal pain was assessed based on the Nordic questionnaire identifying nine possible pain points in the body ."

True Entities
{type: disease, start: 0, end: 2} {type: method, start: 7, end: 9} {type: disease, start: 12, end: 13} {type: region, start: 16, end: 17}
True Relations
{type: help_diagnose, head: 1, tail: 0}

Predicted Entities
{type: disease, start: 0, end: 2} {type: method, start: 7, end: 9} {type: disease, start: 12, end: 13} {type: region, start: 16, end: 17}
Predicted Relations
{type: help_diagnose, head: 1, tail: 0} {type: located_in, head: 2, tail: 3}

Table 9: True vs predicted entities and relations for Sample B.

From 9, we see that the model predicts an additional relation that that appears valid since the pain points are located in the body. However, this was not listed in the initial annotation.

4.4 Cross-Domain Results

Finally, we evaluated cross-domain performance by training on the combined ADKG and MDKG datasets and testing exclusively on ADKG. The corresponding results are reported in Tables 10 and 11, covering NER and RE respectively.

NER had high a F1 scores > 70 for mutation(73.33) and low scores < 50 for other (23.61) and method (43.71). RE had no high F1 scores > 70 but low scores < 50 for all other relations except abbreivation_for. Most noteably, despite having the full ADKG and MDKG type list, ADKG only matched with ADKG related types for both entities and relations.

5 Discussion

5.1 Model Training and Benchmarking

The results are mostly consistent with the exploratory data analysis. Both ADKG and MDKG exhibit class imbalance, with a small number of entity and relation types dominating the distributions. This imbalance is reflected in the model performance, where frequent categories such as the disease entity and abbreviation_for relation achieve higher F1 scores. Rare entities such as other for ADKG and symptom for MDKG had low F1 scores and rare relations such as treatment_target_for for ADKG and located_in for MDKG had low F1 scores.

Table 10: NER with ADKG and MDKG on ADKG by Entity.

Entity Type	Precision	Recall	F1
gene	69.08	64.73	66.84
method	41.80	45.81	43.71
drug	72.25	51.55	60.17
disease	80.60	81.07	80.83
mutation	75.00	71.74	73.33
other	34.62	17.91	23.61
symptom	0.00	0.00	0.00
physiology	0.00	0.00	0.00
signs	0.00	0.00	0.00
region	0.00	0.00	0.00
health_factors	0.00	0.00	0.00
micro	65.90	63.60	64.73
macro	33.94	30.26	31.68

Table 11: RE with ADKG and MDKG on ADKG by Relation.

Relation Type	Precision	Recall	F1
abbreviation_for	62.90	66.42	64.61
help_diagnose	17.95	13.46	15.38
located_in	0.00	0.00	0.00
associated_with	8.51	7.45	7.95
characteristic_of	19.23	10.42	13.51
risk_factor_of	25.81	27.12	26.45
hyponym_of	36.89	40.86	38.78
treatment_for	25.81	20.25	22.70
occurs_in	0.00	0.00	0.00
treatment_target_for	25.00	11.11	15.38
micro	36.99	35.50	36.23
macro	22.21	19.71	20.48

There are some exceptions. For example, the mutation entity is rare in ADKG but achieved a high F1 score. This is likely due to the fact that mutation exhibits unique structural patterns that the other entities do not have.

Error analysis showed that overlapping entities increased error rates. Relation distances also increased error rates. Thus, performance gaps arise from overlapping entities and long-range dependencies. We also observed several edge cases where the predictions appear valid but are not in the ground-truth annotations. Thus, many errors do not come from model failure but instead from consistencies in the annotation process, such as tokenization issues or missed relationships. Therefore, the model is underestimating.

In the cross-domain setting, we incorporated MDKG as auxiliary data and trained ADKG and MDKG but tested on only the ADKG dataset. Although the model had exposure from more diverse group of types, this did not change ADKG results substantially. Thus, simply merging the datasets is not sufficient for effective domain transfer learning.

Overall, performance is largely driven by class imbalance, with frequent categories achieving higher F1 scores than rare ones. Structural challenges such as overlapping entities and long-range dependencies further degrade performance. Some errors also stem from annotation inconsistencies. Finally, the cross-domain results indicate that simply combining datasets is insufficient for effective cross-domain transfer learning.

5.2 Limitations and Future Work

This study is limited by the use of a given model architecture (SpERT) and predetermined training settings. Future work could investigate different model architectures and hyperparameter configurations. Addressing annotation inconsistencies and improving cross-domain transfer learning can also be pursued.

References

Markus Eberts and Adrian Ulges. Span-based joint entity and relation extraction with transformer pre-training. *arXiv preprint arXiv:1909.07755*, 2019.